

The Full Set of Significant Single-Edit Swings

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Abstract—Companion artifact to “Rephrase Before You Act: Characterizing and Mitigating Language Sensitivity in Vision-Language-Action Models”, listing every significant single-edit swing set the searches found, in the paper’s format: differing tokens are highlighted, green marks the best phrasing, and each set carries its swing and exact p. For π_0 on SIMPLER Bridge, the search evaluated 105 candidate pairs and 66 survived a two-proportion z test ($z \geq 1.96$), forming 35 sets over 16 tasks (success rates over $n = 72$ rollouts per phrase; basket task $n = 180$). For $\pi_{0.5}$ on LIBERO, the search evaluated 426 candidate pairs and 115 were significant ($|\Delta| \geq 18$ pp, $p < 0.05$), forming 23 sets; eight were removed as scene-ambiguous — a generalized noun matched more than one object in the scene and the ambiguous phrasing was the one that lost — leaving the 15 sets over 13 tasks listed here ($n = 50$ per phrase). The same records in machine-readable form, one row per phrase, are the two CSV files under `data/` in this repository.

I. SINGLE-EDIT SWINGS FOR π_0 + SIMPLER BRIDGE

Source noun synonym

eggplant_on_sponge +44 pp · $p < 0.0001$
71% set the purple eggplant on the yellow and green sponge
26% set the purple thing on the yellow and green sponge
coke_can_on_keyboard +42 pp · $p < 0.0001$
46% Put the red soda cylinder on top of the black keyboard.
4% Put the red soda can on top of the black keyboard.
pepsi_on_plate +39 pp · $p < 0.0001$
71% set the pepsi on the plate
32% set the soda on the plate
carrot_on_ramekin +38 pp · $p < 0.0001$
47% Pick up the orange carrot and place it in the white bowl.
10% Pick up the orange vegetable and place it in the white bowl.
coke_can_on_keyboard +32 pp · $p < 0.0001$
32% set the coke on top of the keyboard
0% set the can on top of the keyboard

Destination noun synonym

coke_can_on_ramekin +42 pp · $p < 0.0001$
74% place the red cola can inside the white container
69% place the red cola can inside the white bowl
69% place the red cola can inside the white pot
69% place the red cola can inside the white basin
51% place the red cola can inside the white dish
50% place the red cola can inside the white cup
32% place the red cola can inside the white ramekin
coke_can_on_plate +32 pp · $p = 0.0001$
67% put the coke on the plate
61% put the coke can on the plate
53% put the coke can on top of the plate
44% put the coke can on the yellow plate
42% put the coke can on the dish
35% put the can on the plate
carrot_on_wheel +25 pp · $p < 0.0001$
31% put the carrot on the black wheel
12% put the carrot on the wheel
6% put the carrot on the tire

6% put the carrot on the rim
coke_can_on_plate +19 pp · $p = 0.0198$
61% put the coke can on the plate
42% put the coke can on the dish
carrot_on_keyboard +18 pp · $p = 0.0030$
25% place the carrot on the keyboard keys
25% put the carrot on the black keyboard
7% put the carrot on the keys

Preposition

carrot_on_ramekin +20 pp · $p = 0.0111$
40% Place the carrot inside the ramekin.
21% Place the carrot in the ramekin.
pepsi_on_plate +15 pp · $p = 0.0424$
36% Move the blue can onto the plate.
21% Move the blue can to the plate.

Color — addition helps

eggplant_on_sponge +61 pp · $p < 0.0001$
69% purple eggplant goes on the sponge
8% eggplant goes on the sponge
eggplant_on_keyboard +42 pp · $p < 0.0001$
54% put the eggplant on the black keyboard
12% put the eggplant on the keyboard
pepsi_on_plate +38 pp · $p < 0.0001$
64% set the blue coke on the plate
26% set the coke on the plate
cube_on_plate +35 pp · $p < 0.0001$
79% put the green block on the plate
44% put the block on the plate

coke_can_on_wheel +28 pp · $p < 0.0001$
36% Grab the red Coke can and set it on the black wheel.
8% grab the red coke can and set it on the wheel
coke_can_on_wheel +21 pp · $p = 0.0013$
29% grab the red coke can and set it on the black wheel
8% grab the red coke can and set it on the wheel

put_eggplant_in_basket +14 pp · $p < 0.0001$
96% put the eggplant in the yellow basket
82% put the eggplant in the basket

Color — addition hurts

cube_on_plate +30 pp · $p < 0.0001$
85% Pick up the green cube and put it on the plate.
54% pick up the green cube and put it on the yellow plate

eggplant_on_keyboard +30 pp · $p = 0.0002$
61% The eggplant needs to end up on the black keyboard.
31% The purple eggplant needs to end up on the black keyboard.

spoon_on_towel +24 pp · $p = 0.0044$
67% put the spoon atop the towel
61% put the spoon on the towel
44% put the spoon on the blue towel
43% put the spoon atop the blue towel

cube_on_plate +22 pp · $p = 0.0051$
76% pick up the green cube and put it on the plate
54% pick up the green cube and put it on the yellow plate

nut_on_plate +18 pp · $p = 0.0193$
40% drop the nut on the yellow plate
22% drop the grey nut on the yellow plate

cube_on_plate +18 pp · $p = 0.0151$
82% Place the green block on the plate.
64% Place the green block on the green plate.

Action verb

cube_on_plate +38 pp · p<0.0001

78% place the green cube into the dish

40% put the green cube into the dish

eggplant_on_keyboard +24 pp · p=0.0017

40% Rest the purple eggplant on the black keyboard.

17% Move the purple eggplant on the black keyboard.

eggplant_on_sponge +19 pp · p=0.0051

88% Set the purple eggplant onto the sponge.

68% Move the purple eggplant onto the sponge.

nut_on_wheel +17 pp · p=0.0071

25% Put the nut right on the wheel.

8% Position the nut right on the wheel.

Phrase structure

eggplant_on_sponge +49 pp · p<0.0001

85% Set the purple eggplant on the sponge.

36% purple Eggplant set on the sponge.

pepsi_on_plate +44 pp · p<0.0001

69% put the pepsi on the plate

25% put the Pepsi on the plate

pepsi_on_plate +43 pp · p<0.0001

69% set the pepsi down on the plate

26% Set the Pepsi down on the plate.

pepsi_on_plate +39 pp · p<0.0001

69% put the pepsi on the plate

31% Put the Pepsi on the plate.

eggplant_on_keyboard +19 pp · p=0.0051

32% put eggplant on keyboard

12% put the eggplant on the keyboard

Multi-edit

stack_cube +33 pp · p<0.0001

50% put the green cube on the yellow cube

43% green cube on yellow cube

29% put the green block on the yellow block

25% stack the green cube on the yellow cube

21% stack the green block on the yellow block

17% green block on yellow block

II. SINGLE-EDIT SWINGS FOR $\pi_{0.5}$ + LIBERO

Noun synonym

libero_goal/7 +100 pp · p<0.0001

100% switch on the electric stove

100% switch on the stove

100% switch on the stovetop

100% start the stove

100% power on the stove burner

86% switch on the range

86% turn the black knob to start the burner

84% switch on the cooker

84% turn the black knob to activate the burner

78% switch on the hob

74% switch on the cooktop

66% switch on the heating element

64% switch on the burner

62% switch on the griddle

48% start the burners

40% start the burner

30% switch on the hotplate

2% switch on the hot plate

2% switch on the electric hot plate

0% power on the hot plate

libero_goal/5 +98 pp · p<0.0001

98% Go ahead and push the plate right up to the front

of the stove.

98% push the plate to the front of the stove

62% slide that plate over to the front of the stove

2% slide that dish over to the front of the stove

0% Slide that dish over to the front of the stove.

0% slide that flat dish over to the front of the

stove

libero_goal/1 +92 pp · p<0.0001

92% place the grey bowl on the electric burner

18% place the grey bowl on the hotplate

0% place the grey bowl on the electric hot plate

0% place the grey bowl on the hot plate

libero_spatial/7 +78 pp · p<0.0001

100% pick up the black bowl on the stove and place it on the plate

22% pick up the black bowl on the hot plate and place it on the plate

libero_goal/8 +64 pp · p<0.0001

100% put the bowl on the plate

100% put the bowl on the plate.

48% put the bowl on the dish

36% put the bowl on the flat dish

libero_90/30 +56 pp · p<0.0001

76% put the black bowl on the plate

20% put the black bowl on the dish

libero_spatial/1 +38 pp · p<0.0001

100% pick up the black bowl next to the ramekin and place it on the plate

62% pick up the black bowl next to the dish and place it on the plate

libero_object/2 +26 pp · p=0.0001

100% pick up the salad dressing and place it in the basket

100% pick up the dressing and place it in the basket

74% pick up the bottle and place it in the basket

Color — addition helps

libero_goal/7 +46 pp · p<0.0001

86% turn the black knob to start the burner

40% start the burner

Color — addition hurts

libero_90/31 +26 pp · p=0.0020

36% put the bowl on top of the cabinet

16% put the black bowl on top of the cabinet

10% put the grey bowl on top of the cabinet

libero_goal/9 +26 pp · p=0.0071

76% put the wine bottle on the stand

50% put the wine bottle on the wooden stand

Action verb

libero_90/44 +94 pp · p<0.0001

100% fire up the stove

100% ignite the stove

100% power on the stove

100% start the stove

100% switch on the stove

6% turn on the stove

libero_goal/7 +60 pp · p<0.0001

100% turn on the burner

100% turn on the burners

64% switch on the burner

48% start the burners

40% start the burner

libero_90/38 +20 pp · p=0.0440

54% pick up the right moka pot and place it on the stove

34% lift the right moka pot and place it on the stove

Preposition

libero_90/10 +40 pp · p<0.0001

100% place the black bowl on top of the cabinet

60% place the black bowl up onto the cabinet